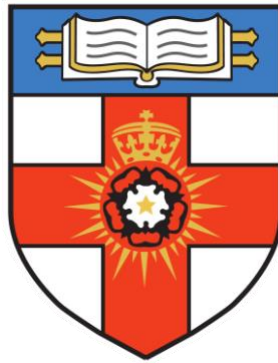


UNIVERSITY OF LONDON

INTERNATIONAL PROGRAMMES

BSc Computer Science and Related Subjects



CM3070 FINAL PROJECT

PRELIMINARY PROJECT REPORT

Little Legends: Children's Storybook AI Generator

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INTRODUCTION

PRIMARY CONCEPT – *LITTLE LEGENDS: CHILDREN STORYBOOK AI GENERATOR*

Little Legends is an AI-powered storytelling platform (web app) that allows preschool children to become the heroes of their own adventures. By combining a child's self-drawn image, and short prompt words or phrases, the system generates a personalized, illustrated story — even reading it aloud to the child. The project orchestrates multiple pre-trained AI models across different modalities (image, text, and audio) to build a fun, educational tool, tailored to non-literate children's imaginations.

The project addresses the challenge of blending visual, linguistic, and audio-based machine learning in a cohesive, real-time storytelling experience for young users and their guardians/educators.

INTERPRETATION AND TAILORING OF TEMPLATE 4.1

Based on the template “**4.1 Orchestrating AI Models to Achieve a Goal,**” this project integrates multiple AI models—such as image captioning, keyword extraction, natural language generation, text-to-speech, and image synthesis—in a coordinated pipeline to generate personalized children's storybooks from hand-drawn images and text prompts.

Figure 1 shows the multimodal pipeline involved in Little Legends that turns real-world input (drawings, prompts) into interactive narrative output (text + speech), demonstrating its applications in the domain of early childhood education, creative play and digital inclusion.

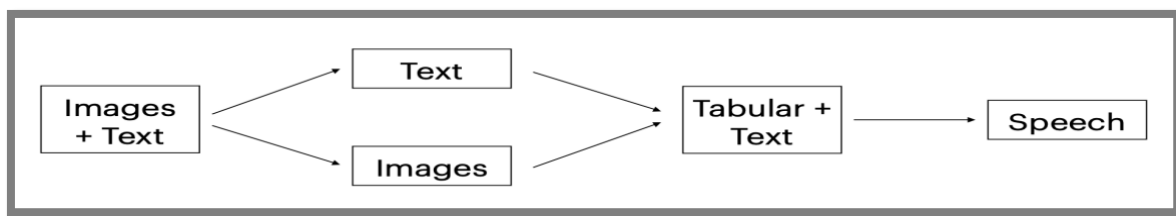


Figure 1 - Overview of the multi-modal pipeline of Little Legends

The final system integrates **three core** AI model capabilities¹:

The **Drawing Analysis** phase begins with an image model tasked with identifying and classifying key objects in the child's sketch. This model performs image captioning by interpreting visual elements—such as a dragon, castle, or tree—and translating them into descriptive textual phrases that can be understood by downstream language models.

Next, **Keyword Extraction** is carried out by a text model, which processes any accompanying text prompt provided by the user. It identifies the most relevant and meaningful words, typically nouns or actions, that can serve as anchors for narrative development. These keywords are then combined with those derived from the drawing to inform the story's direction.

In the **Language Generation** stage, a text generation model (often using prompt engineering techniques) constructs the actual story. It integrates the child's name, any user-provided prompts, and visual elements extracted from the drawing to craft a cohesive and imaginative narrative suitable for a young audience.

The **Voice Narration** component utilizes a text-to-speech (TTS) model to convert the written story into audio. This narration is designed to be expressive and child-friendly, enhancing accessibility and engagement, especially for pre-literate children or those with learning differences.

To unify visual and textual understanding, a simple **Data Flow** mechanism combines both sets of keywords into a single concept list. This consolidated input serves as the blueprint guiding the language model's generation of the story, ensuring that both visual and verbal cues are harmoniously represented (e.g., combining "dragon," "princess," and "adventure" into a narrative plotline).

Finally, an **Optional Enhancement** step can bring the experience full circle. By using an image generation model, the child's drawing is transformed into vibrant illustrations that match scenes or characters from the story. These visuals are then compiled with the narrated story to produce a multimedia digital book—complete with text, images, and audio—for a personalized and immersive storytelling experience.

¹ Refer to **Chapter 3: Project Design** for more information on the pre-trained models used in the project.

MOTIVATION

Children between the ages of 3 and 6 are at a crucial stage of cognitive, linguistic, and emotional development. Storytelling fosters imagination, boosts verbal comprehension, and helps children develop empathy and confidence. It plays a powerful role in developing early literacy, imagination, and emotional expression in young children. According to Philips (1999)², engaging with stories helps children build language skills while also allowing them to explore feelings and ideas creatively.

Additionally, when it comes to parents or educators, even when storytelling doesn’t come naturally to them, they still play a crucial role in nurturing this skill. Research from the *European Early Childhood Education Research Journal* (2012)³ highlights that adult involvement—through encouragement and shared story experiences—significantly supports children’s storytelling development.

However, most AI storytelling tools are static or adult-led, often lacking features that centre the child’s input. Many do not support uploading of children’s drawings, rely on adult-oriented interfaces, or animate artwork without offering accompanying text or narration. *Little Legends* addresses these gaps by empowering children to co-create their own stories using simple tools: their drawings, keywords, and imagination.

RELATED WORKS & RESEARCH

COMPARISON OF RELATED PROJECTS⁴

Project	Strengths	Limitations	Gaps
Storybee	Integrates narrative, visuals and audio Customizable characters & themes	No input from drawings or voice Geared towards literate users	Incorporates children’s drawings and spoken prompts to improve accessibility for pre-literate children

² Phillips, Louise. (1999). The Role of Storytelling in Early Literacy Development.
³ Umek, L. M., Fekonja-Peklaj, U., & Podlesek, A. (2012). Parental influence on the development of children’s storytelling.
⁴ Refer to **Appendix A** for more details on each related project.

Bedtimestory AI	Fast and personalized Simple, fun interface	Text-only input Adult-centric design No visual input from children	Introduce drawing-based input for visual co-creation and better child engagement
Drawings Alive	Animates children’s artwork Highly engaging and visual Encourages creativity	No storytelling component No narration Limited to visual animation	Extend functionality to generate narrative content based on drawings and keywords, integrating text and speech
Dashtoon’s AI Children’s Book Generator	High-quality illustrations Easy-to-use interface Personalized stories	Requires typed input No drawing or voice input No audio narration	Add multimodal inputs (drawings, voice) and audio storytelling to enhance accessibility and interactivity

Figure 2 - Comparison of Related Works

A critical comparison in **Figure 2** of AI storytelling platforms reveals that while they offer features like personalization, animation, or illustrations, they fall short in supporting fully child-centred, multimodal interaction. Most rely on text input and adult facilitation, limiting access for pre-literate children. Even tools like Drawings Alive, which allow visual input, lack narrative generation and voice support.

Little Legends addresses these limitations by generating fully personalized and narrated stories using children’s drawings and keywords. Furthermore, emerging research has demonstrated that AI tools can significantly support early childhood development, enhancing literacy, imaginative expression, emotional self-regulation, and collaborative learning among young children (Su & Yang, 2022)⁵. When combined with storytelling, AI can further enrich learning experiences by making narratives more engaging, interactive, and accessible.

⁵ Su, J., & Yang, W. (2022). Artificial intelligence in early childhood education: A scoping review.

LITERATURE REVIEW

MULTIMODAL STORYTELLING SYSTEMS

EMERGING TECHNOLOGIES AND IMPACT

The advent of advanced multimodal AI technologies has enabled transformative applications in educational and storytelling domains, significantly impacting early childhood learning experiences. Recent developments in distinct modalities such as image-to-text, text generation, text-to-speech, and image generation demonstrate critical implications for innovative storytelling platforms (Bensaid et al., 2023), introducing automated storyline decision-making in generated stories with little to no human intervention (Kim et al., 2023).

Significant advancements have emerged through the use of state-of-the-art pre-trained AI models—particularly transformer-based language models and diffusion models for image synthesis (Kim et al., 2023). For instance, OpenAI's GPT models have been extensively studied for their ability to extract semantic cues from both text and image inputs to generate coherent narratives (Brown et al., 2020). These models are often integrated with BERT in storytelling pipelines to enhance contextual understanding and content alignment (Chen, 2021; Kim et al., 2023).

To take things another step further, recent work in video diffusion and animation interpolation has opened new avenues for bringing static illustrations to life through short animated sequences. Techniques such as segment-guided video interpolation (Li et al., 2021) and sticker-style motion diffusion (Yan et al., 2024) allow child-drawn or AI-generated images to be transformed into engaging animations while preserving artistic intent and character identity. These innovations can enhance narrative immersion, particularly in children's storybooks, by turning illustrated scenes into vivid, animated segments that support comprehension, emotional engagement, and sustained attention.

These multimodal input and output AI-driven storytelling platforms has shown pedagogical impact on the language acquisition, vocabulary development and literary outcomes in young children exposed to them in settings like Asia and China (Chen, 2024).

TECHNOLOGY

NATURAL LANGUAGE GENERATION – MAINTAINING NARRATIVE COHERENCE

Recent advances in neural text generation have delivered impressive fluency and creativity, thanks to transformer architectures like GPT-3/4 and BERT-based variants (Zheng et al., 2024). However, long-form narrative coherence—ensuring that generated stories remain logically consistent across sentences and maintain character continuity—remains a core challenge (Bensaid et al., 2023).

For instance, Slobodianiuk and Semerikov (2025) systematically reviewed state-of-the-art neural models and noted that while embedding-based metrics (e.g., BERTScore) improved evaluation, human assessments are still essential for judging coherence. Other work focuses specifically on story coherence, such as approaches that track entities, events, and reader-model feedback.

Clark et al. (2018) attempted to enhance coherence by embedding explicit representations of characters and entities, helping models maintain narrative consistency, whereas Coherence-boosting strategies like generating skeleton outlines before full sentences was demonstrated by Xu et al. (2018)—showing reduced repetition and improved cohesion.

Evaluation remains a challenge. Alongside BLEU and ROUGE, novel LM-based evaluation techniques now assess narrative quality holistically—scoring aspects like consistency, plot progression, and thematic relevance. In the context of our project—generating child-friendly stories from drawings and prompts—it’s vital to adopt models that not only understand input context but sustain coherent, engaging narratives.

IMAGE GENERATION MODELS & CHALLENGES – ACCURACY & PRECISION

Diffusion-based architectures such as DALL·E 2 and Stable Diffusion have redefined the capabilities of text-to-image generation by leveraging denoising diffusion probabilistic models (DDPMs) to iteratively reconstruct high-fidelity images from noise (Ramesh et al., 2022). These models excel at generating visually rich and stylistically coherent illustrations based on textual prompts, offering fine-grained control over visual composition and semantics.

Despite these strengths, maintaining accuracy and semantic precision in generated outputs remains challenging—particularly in educational or child-centric contexts. A common limitation is prompt-image misalignment, where key elements in the input prompt are omitted or misinterpreted. This issue becomes more pronounced with abstract or imaginative prompts from children, which require higher-level comprehension and creative generalization. Saharia et al. (2022) emphasize that even top-performing models encounter difficulty in depicting complex scenes involving multiple entities or spatial relationships.

Evaluation frameworks often rely on quantitative metrics like Fréchet Inception Distance (FID) for realism and CLIPScore for semantic congruence, yet these alone are insufficient for assessing appropriateness in child-oriented media (Hessel et al., 2022).

Moreover, concerns persist around biased datasets, restrictive content filters, and resource-heavy inference. Image models trained on unbalanced data may reinforce harmful stereotypes, and safety mechanisms—designed to block inappropriate content—can mistakenly censor innocent child-generated imagery (Yang, 2025). Addressing these issues requires curated training datasets, nuanced filtering systems, and prompt-engineering methods that are sensitive to the needs of young audiences.

SOCIAL IMPACT

AI IN EARLY CHILDHOOD EDUCATION & CHILD-COMPUTER INTERACTION

AI is increasingly being explored as a transformative tool in early **childhood education (ECE)** (Chen, 2024); particularly in the areas of storytelling, language acquisition, and creative expression. It can aid educators in reducing time and costs while fostering creativity in children (Tengku Hariffadzillah et al., 2023).

Child-computer interaction (CCI) research explores the need for technology that aligns with children's cognitive, emotional, and motor development stages (Read & Bekker, 2011). As a result, interfaces should ideally be intuitive, voice-supported, and visually rich, minimizing adult intervention with a more child-centric system, giving them more agency at constructing their own learning (Kucirkova et al., 2019), which is what I am hoping for with this project.

However, challenges remain in ensuring age-appropriate content, sustaining attention, and promoting safe and meaningful engagement. Studies suggest that integrating AI with principles from developmental psychology can help scaffold learning and encourage storytelling as a co-creative process rather than a passive experience (Kucirkova et al., 2019). However, the body of research on AI's role in early childhood education remains limited (Su & Yang, 2022). Continued long-term investigation is essential to fully harness AI's potential in preparing the next generation for an increasingly digital future (Chen, 2024).

BIAS IN LANGUAGE & IMAGE MODELS

AI-generated stories and illustrations can unintentionally reflect harmful biases from their training data, subtly reinforcing stereotypes in visual storytelling. Recent studies have shown that even in seemingly neutral contexts like children's bedtime stories, large language models can produce narratives that reflect gender and racial bias. For example, prompts like *"Write a bedtime story about a child who grows up to be a {occupation}. Once upon a time"* revealed disparities in how different social identities were portrayed, with characters' competence, agency, and narrative roles varying based on their race or gender (Lum et al., 2024).

A Nature article (2024) reports that large language models (LLMs) exhibit social identity biases akin to human in-group favoritism and out-group derogation, though careful fine-tuning and data curation can mitigate these tendencies (Hu et al., 2025). Meanwhile, Kotek et al. (2023) showed that LLMs still reinforce harmful gender stereotypes, even when tuned with human feedback.

As for Image generation bias, a recent open-access study by Yang (2025) found that AI-generated images frequently default to depicting White individuals more accurately than people of color, especially in face-swapping contexts, indicating systemic racial bias in image-to-image systems. Similarly, Castleman and Korolova (2025) demonstrated persistent "adultification bias" in both text-to-image and language models, which disproportionately depict Black girls as older and more sexualized than their White counterparts.

These findings underscore the ethical risks of using multimodal AI in story generation. They highlight the importance of prompt engineering, bias mitigation strategies, and careful model selection—especially when creating stories for children, where representational fairness and safety are paramount.

PROJECT DESIGN

DOMAIN & USERS

The design of *Little Legends* is grounded in the intersection of early childhood education, creative storytelling, and assistive AI technologies. The primary users—**preschool-aged children**—are at a developmental stage characterized by emerging symbolic thinking, limited literacy, and a strong affinity for visual and auditory stimuli. The domain of the project—**storytelling for early education and emotional development**—requires tools that are accessible and supportive of imagination without placing cognitive strain on the user.

To meet these needs, the system emphasizes **visual input over text**, allowing children to express ideas through hand-drawn images. This removes literacy as a barrier and leverages drawing, a developmentally appropriate medium, as the primary mode of interaction. By incorporating **AI-generated narratives** from these drawings and minimal textual prompts, the project supports children's narrative thinking and language acquisition in a way that feels natural and playful.

The system also generates **visual AI illustrations** in response to the child's drawing and prompt, reinforcing comprehension through visual storytelling while offering children a tangible sense of creative co-ownership. To further accommodate early learners, the integration of **speech synthesis** allows the stories to be read aloud—supporting non-readers and enhancing listening comprehension and narrative engagement. This feature also benefits **parents and educators who may find storytelling creatively challenging or feel shy expressing ideas**, despite having a strong desire to inspire and connect with children. By offloading the pressure of performance and invention, the system empowers adult facilitators to participate more confidently in co-creative storytelling moments.

Technically, the system avoids complex input flows or overloaded interfaces, opting instead for a **minimal, modular UI** that caregivers or educators can help mediate if necessary.

In short, every design decision—from the input mechanism to the multimodal outputs—has been chosen to create a system that is age-appropriate while satisfying the domain requirements of supporting language development, creativity, and educational storytelling.

SYSTEM ARCHITECTURE & TECHNICAL APPROACH

SYSTEM ARCHITECTURE

1. Frontend Interface (React Web App)

The frontend of *Little Legends* is a React-based web application crafted to be simple, accessible, and creatively stimulating for children and their caregivers. The interface allows users to upload a child's drawing (`<input type="file" onChange={handleFileChange} />`) and input a short keyword or phrase (`<input type="text" value={prompt} onChange={handlePromptChange} />`). Upon clicking "Generate Story," the React component constructs a FormData payload and sends it via a `fetch()` POST request to an Express backend server. Accessibility is supported via the browser's Web Speech API (`window.speechSynthesis`) at this moment of development, but will be replaced with a better AI model after more testing.

2. Node.js Backend (Express Server)

The backend server, implemented in Node.js using Express, serves as the orchestration layer that connects user input to AI-generated outputs. When the React frontend sends a POST request to the `/generate` endpoint, the server receives the uploaded drawing and accompanying prompt using `multer`

middleware (upload.single('drawing')). This method utilizes the *gpt-4.1-nano model* to generate a narrative from both textual and visual inputs.

Once the story is generated, the image buffer is written to a local input directory which is synced with Google Drive via the filesystem path. The server then enters a polling loop to check for the availability of the AI-generated image, which is produced by a separate Python process on Google Colab, utilising a combination of pre-trained AI models from Hugging Face for image generation. This asynchronous coordination enables real-time feedback while maintaining loose coupling between the image generation pipeline and the main server.

3. Colab AI Processing (Python)

At this development stage, the project is using the Colab-based AI pipeline with a ControlNet-enhanced Stable Diffusion model to conditionally generate high-quality illustrations based on a child’s drawing and textual prompt. Implemented using Python and the HuggingFace diffusers library, the system loads the *Illiasviel/control_v11p_sd15_scribble model* in tandem with *runwayml/stable-diffusion-v1-5*. Input images are monitored in a Google Drive–synced folder, processed in real time, and resized for optimal efficiency. The generated illustrations are saved to a designated output folder in Drive, allowing seamless retrieval by the Express.js server for delivery to the frontend interface.

TECHNOLOGY & PRE-TRAINED AI MODELS			
Modality	Technology or AI Model to consider	Reasoning	Approach
Drawing Analysis + Image Captioning (Image Model)	- OpenAI GPT-4.1 nano (prototype – for testing pipeline) - BLIP-2 (via Hugging Face or MiniGPT-4 with BLIP2)	- BLIP-2 is lightweight to run in Colab - Strong at both captioning and concept detection - Free and open-source	Model with higher image analysis accuracy will be used.

Image Generation (Image-to-Image / text-to-image Model)	- DALL-E 3 by OpenAI (ideally – but too costly) - Stable Diffusion (v1.5/v2.1/XL) + ControlNet – Hugging Face (prototype – but may use this if test results are good)	- Open-source (Stable Diffusion) - Free to use locally or on Colab - Good-enough quality images - Customizable via prompt engineering/fine-tuning	Will most likely choose the more cost-effective model if the image quality are acceptable.
Keyword Extraction (Text Model)	KeyBERT	- Ideal for short prompts - Understands semantic meaning - Suitable for child-like input	Will have to test to see if this is necessary.
Language Generation (prompt engineering) (Text Generation Model)	- OpenAI GPT-4.1 nano (prototype – for testing pipeline) – story generation from image + text prompt - MiniGPT-4	- Already purchased OpenAI's GPT-4.1 nano for prototype testing, stories generated are decent. - MiniGPT-4 is free and open-source, colab-supported	MiniGPT-4 might be more complicated to setup compared to GPT-4.1 nano, in that case may continue with GPT-4.1 nano.
Voice Narration (Text-to-Speech Model)	- Web Speech API – speechSynthesis (prototype – for testing pipeline) - GPT-4o-mini-TTS (can be costly)⁶ - Coqui TTS - XTTS v2 (by Coqui) - VITS (Variational Inference TTS)	- Coqui TTS or XTTS v2 are natural and cheerful-sounding - XTTS v2 allows cloning of storyteller's voice - Both are free and open-source - VITS is more resource-intensive; setup is harder than Coqui	Will test to see which model is sounds most suitable and cost-effective

⁶ Refer to **APPENDIX C** to see details on GPT-4o-mini-TTS model.

Data Flow between Models – Content Understanding (Concept List) Simple Python Code	• Google Colab (Python) • Gradio (for quick AI model testing)	• Google Colab to process the AI image generation – free access to a GPU for testing POC & integrates with Google Drive • Gradio to test the image generation model	Will continue testing to see if this combination works with the full-stack development.
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Figure 3 - Analysis of Pre-trained AI models & Technology

To bring the vision of personalized, engaging children’s stories to life from simple drawings and prompts, this project strategically integrates a curated selection of AI technologies—outlined in **Figure 3**—that balance creativity, technical feasibility, and cost-efficiency. For interpreting children’s artwork and generating descriptive captions, models such as BLIP-2—available through Hugging Face—or MiniGPT-4 are being explored for their strong visual reasoning capabilities and seamless deployment within environments like Google Colab. Story generation is currently handled by OpenAI’s GPT-4.1 nano for prototyping due to its high-quality outputs, while MiniGPT-4 is being considered as a sustainable, open-source alternative for long-term use.

To generate accompanying illustrations, Stable Diffusion enhanced with ControlNet is employed for its adaptability and image quality, providing a viable, cost-free substitute to more expensive options like DALL·E 3. The system may also incorporate KeyBERT for keyword extraction, especially useful for understanding brief or imaginative prompts written by children.

Voice narration is initially handled through the browser-native Web Speech API for accessibility and prototyping purpose, with more natural and expressive solutions like Coqui TTS or XTTS v2 under consideration to enhance the auditory storytelling experience. Processing and orchestration are done within Google Colab to take advantage of free GPU access, and Gradio is available for quick model testing during development.

Altogether, this architecture aims to find an optimal balance of system performance, scalability, and creative flexibility—leveraging accessible, cost-effective technologies to deliver a responsive, multi-modal storytelling experience.

PROJECT PLANNING

The project planning begins with a clear outline of the **main tasks**⁷, which include finalizing the UI/UX design, completing full-stack development, selecting and integrating the most suitable AI models for each modality (text, image, and voice), conducting user acceptance testing, and writing the final report.

Key milestones (Figure 4) include the completion of backend and frontend integration, AI model validation, successful user testing with early childhood educators and parents, and final report & demo preparation. Buffer periods are built into the schedule to account for potential delays or technical issues, ensuring that critical deadlines can still be met even if contingencies arise.

TIMELINE & MILESTONES

S/N	Task (duration)	Estimated Start Date	Estimated End Date	Dependency & Details
1.	Finalize Wireframe & UI/UX Design (1 week)	16 th June	22 nd June	Research & collate related websites' good UI/UX designs.
2.	Confirm final models for each AI modality (1 week)	23 rd June	29 th June	Test & compare each AI pre-trained model for their respective modality.

⁷ Refer to **APPENDIX B** for more details on the **main tasks** for the project.

3.	Integrate final models into Colab pipeline (1 week)	30 th June	6 th July	Task 2 – AI models confirmation Conduct functional testing to make sure system architecture works.
4.	Improve drawing-to-prompt pipeline & test flow (1 week)	7 th July	13 th July	Task 2 – AI models confirmation Task 3 – Ensure Colab works Conduct functional testing to ensure system architecture works.
5.	Connect & test full-stack pipeline end-to-end (1 week)	14 th July	20 th July	Tasks 1, 2 & 3. Conduct functional testing to ensure system architecture works.
6.	Refine story generation + AI image quality (1 week)	21 st July	27 th July	Task 5 Testing to ensure optimal accuracy and precision.
7.	Integrate voice narration alternatives (TTS testing) (Draft report writing) (2 weeks)	28 th July	10 th August	Task 6 Test out different voice narrations according to specific story generated. Start on report draft.
8.	Conduct UAT with educators/parents (1 week) *(submit report draft)	11 th August	17 th August	Conduct user testing and have users fill out performance feedback form.

9.	Iterations + address issues from UAT evaluations (Incorporate feedback+iterations to finalize full report) (2 weeks)	18 th August	31 st August	Task 8 Evaluate feedback from functional and user testing + make iterations Begin finalization of Final Report
10.	Record and edit demo video (1 week)	1 st Sept	7 th Sept	Make PowerPoint slides + screen-record to make demo video
11.	Buffer / Contingency period (1 week)	8 th Sept	14 th Sept	NA
Final Project Submission Deadline: 15 th Sept 2025				

Figure 4 - Project Timeline & Milestones

PROJECT EVALUATION (METRICS)

TEST-DRIVEN DEVELOPMENT (TDD) – FUNCTIONAL TESTING

The TDD approach ensures each system component behaves as intended before integration. Unit tests are written for the React UI (using Jest and React Testing Library) to verify that file-upload, prompt input, and “Generate Story” actions invoke the correct handlers and update state appropriately. On the backend, tests built with Mocha or Jest mock file uploads and OpenAI responses against the /generate endpoint to confirm valid JSON responses, robust error handling, and proper static file serving. Writing tests up front and running them continuously catches regressions early and maintains a solid foundation for subsequent development.

AI MODELS TESTING – ACCURACY & PRECISION

To evaluate the AI models' performance, both **accuracy** (how well the generated stories and images reflect the child's drawing and prompt) and **precision** (the consistency of high-quality outputs across multiple runs) will be measured along with **F1-scores** to balance precision and recall in the assessments.

For story generation, **benchmark testing** will be conducted by sampling 30 diverse prompts and evaluating each narrative against a detailed rubric that assesses relevance, coherence, and creativity. Then, **precision** (the percentage of rubric criteria met) and **recall** (the proportion of intended story elements covered) are calculated to quantify each model's performance. Additionally, A/B preference tests—presenting side-by-side outputs from GPT-4.1 nano and MiniGPT-4 to educators and parents—are run to collect direct human judgments on which model produces more engaging and appropriate stories.

An **annotated test set** of drawings with known objects will be used to apply accuracy testing (object detection recall) alongside precision testing (false-positive rate) for ControlNet-conditioned Stable Diffusion outputs. **Cross-validation** will be performed by generating multiple images per prompt to quantify consistency. Finally, **qualitative error analysis** will document outlier cases and guide any necessary model fine-tuning or replacement.

Objective and subjective evaluations will be conducted on a representative set of story passages to compare TTS engines. First, 20 sample story snippets—varying in length, vocabulary, and emotional tone—will be fed into each engine (Web Speech API, Coqui TTS, XTTS v2, VITS) to measure latency (time from text input to audio start) and resource usage. Next, a survey administered to parents and educators will collect five-point ratings for naturalness, clarity, and emotional expressiveness. Finally, an A/B preference test will present paired recordings from two models for the same passage, asking respondents to indicate their preferred voice. Combined results from these tests will inform the selection of the most child-friendly, performant, and expressive TTS solution.

USER-ACCEPTANCE TEST (UAT) - USABILITY TESTING

Given the sensitivity of working with preschool children, UAT sessions will involve only **parents, educators, and early-childhood professionals**. A standard questionnaire will capture subjective feedback on ease of use, engagement, and perceived value. By focusing on adult proxies, ethical

constraints are respected while still gathering actionable insights into how the interface supports or hinders the co-creative storytelling process.

FEATURE PROTOTYPE

DESIGN & FEATURES

USER JOURNEY

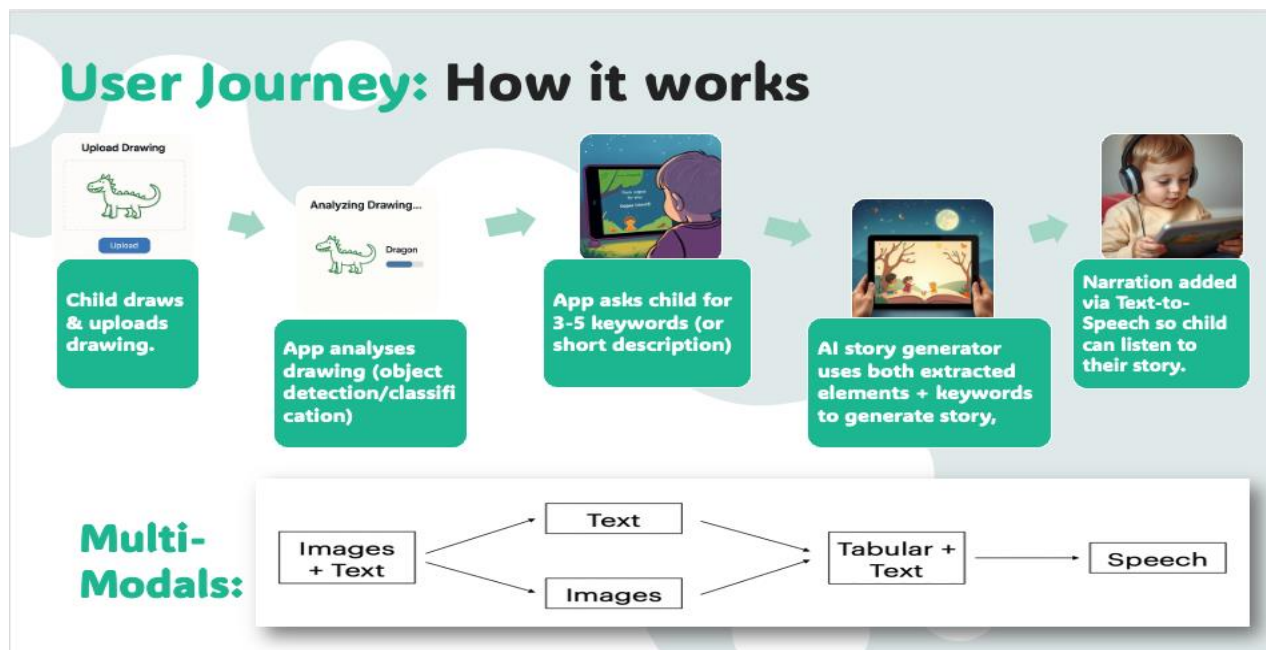


Figure 5 - User Journey of Little Legends

The **user journey** as shown in **Figure 5** begins with a child (or their caregiver) **uploading a hand-drawn image** via the app interface, accompanied by a **simple word or phrase** as a storytelling prompt. Upon submission, the system generates a personalized story using AI, inspired by both the drawing and the text. Simultaneously, a **visual AI illustration** is produced to complement the narrative. Once both outputs are ready, the story is displayed alongside the AI-generated image, with an option to have the story read aloud through **speech synthesis**. This seamless flow—from creation to storytelling to audio playback—ensures an engaging, low-barrier experience that encourages imagination, co-creation, and playful interaction.

WEB APP INTERFACE



Figure 6 - Little Legends Web App Prototype

The current prototype shown in **Figure 6** implements a fully functional upload-and-generate workflow: children can upload their drawing via the “Upload Drawing” button and type a simple prompt, then tap “Generate Story” to kick off AI processing. While the models run, a dynamic “Generating story... Please wait” message with animated dots keeps users informed. Once ready, the AI-written story appears alongside a placeholder (or real) illustration area—falling back to a friendly “Image not available” graphic if the pipeline is still catching up. Text-to-speech controls (Read Aloud, Pause, Stop) let users hear the story, and the clean, softly coloured background and centred card layout ensure a warm, child-friendly interface.

PROTOTYPE EVALUATION

The current prototype implements story generation from multimodal inputs, image uploading & text prompts, and basic TTS narration; although the image-generation pipeline, despite being fully architected and provisioned, is not yet displaying results in the frontend.

STORY GENERATION

The server sends both the drawing (dataUri) and the text prompt (prompt) to the GPT-4.1-nano model, receives the generated narrative in story, and returns it as JSON; the React frontend then submits the drawing and prompt via a fetch() POST request, extracts data.story from the response, stores it in state, and renders it inside the `<p>{story}</p>` element. **(Figures 7 & 8)**

```
63 // Call OpenAI to get the story
64 try {
65   const response = await openai.chat.completions.create({
66     model: "gpt-4.1-nano",
67     messages: [
68       {
69         role: "user",
70         content: [
71           {
72             type: "text",
73             text: `Generate a story for children based on this picture and the following prompt: '${prompt}'.`
74           },
75           {
76             type: "image_url",
77             image_url: {
78               url: dataUri
79             }
80           }
81         ]
82       }
83     ],
84     max_tokens: 500
85   });
```

Figure 7 – Backend: Invoking OpenAI gpt-4.1-nano to generate story

```
41   const formData = new FormData();
42   formData.append("drawing", drawing); // File field
43   formData.append("prompt", prompt); // Text field
44   setIsGenerating(true);
45   try {
46     const response = await fetch("http://localhost:1337/generate", {
47       method: "POST",
48       body: formData,
49     });
50
51     const data = await response.json();
52     setStory(data.story);
53     setAiImageUrl(data.aiImageUrl);
54     setIsGenerating(false);
55   } catch (error) {
56     console.error("Error generating story:", error);
57     setStory("Sorry, something went wrong while generating your story.");
58     setIsGenerating(false);
59   }
60   };
```

Figure 8 - Frontend: Receiving & Displaying Generated Story

However, when testing with a child-drawn wolf sketch (**Figure 9**), the combined drawing-and-prompt approach reliably produces coherent narratives—for example, a simple wolf sketch with the prompt “A wolf with fire mane and tail wants to become my little pony” yields an appropriately themed story (**Figure 10**)—whereas relying on the drawing alone can generate mismatches (e.g., interpreting a wolf sketch as a “blue horse,” as illustrated in **Figure 11**).



Figure 9 - Child-drawn wolf sketch

To address this, a quick captioning step may be required—using a vision-language model like BLIP-2—to convert the sketch into a clear description before it reaches the story generator. Multiple captioning models can be benchmarked on the labelled sketch dataset by measuring precision and recall, and prompt templates plus temperature settings can be tuned to optimize output quality. This additional step provides the story model with a concise summary to work from, reducing odd mismatches.

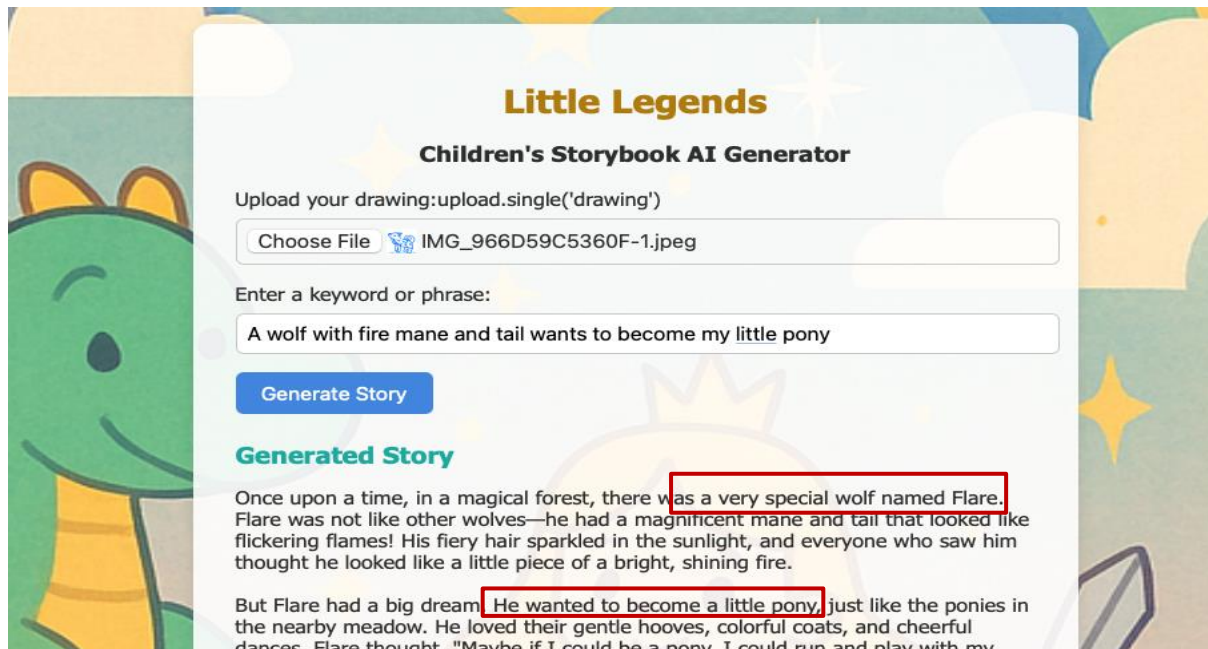


Figure 10 - Image + Prompt generated story

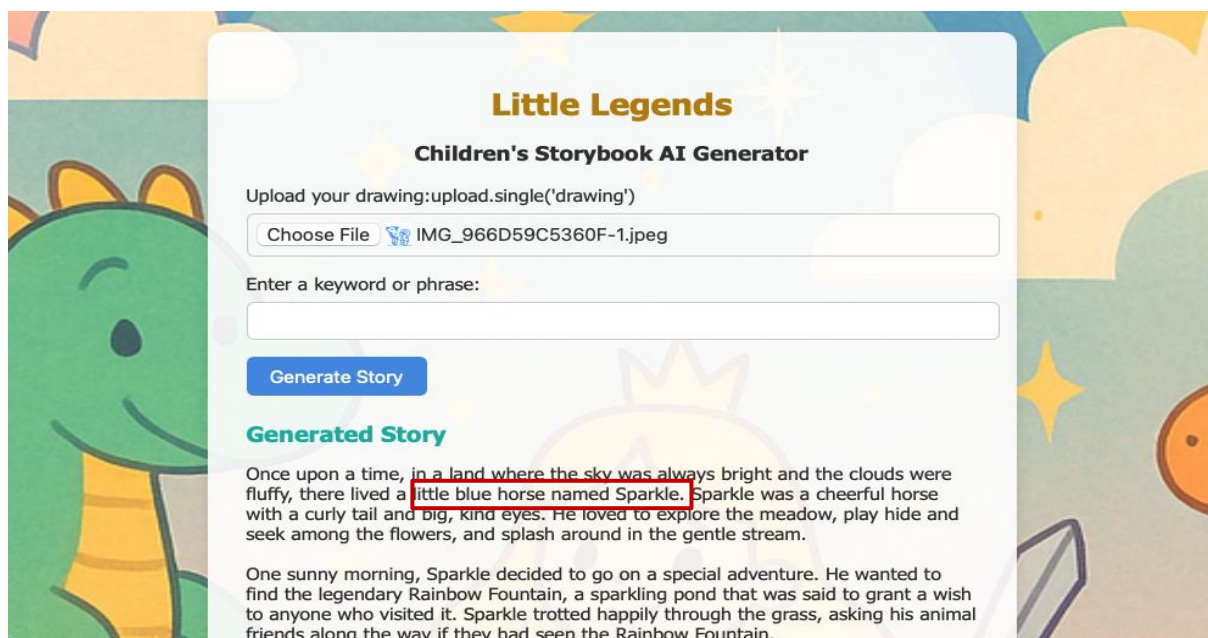


Figure 11 – Image-only generated story

IMAGE GENERATION

The image-generation pipeline, managed through a Colab script and Google Drive syncing (**Figure 14**), is encountering integration challenges: although input files are correctly saved and processed, the resulting illustrations often fail to display in the frontend and occasionally return blank canvases or NSFW warnings (**Figure 12 & 13**)—likely due to GPU constraints and model misconfigurations.

To resolve these integration hiccups, Google Drive sync must be verified to be real-time and that the backend's polling interval is sufficient. Next, Colab logs will be inspected for CUDA or memory errors and, if necessary, I will switch to a smaller or CPU-compatible model variant to eliminate GPU timeouts. To prevent blank outputs and NSFW triggers, safety checker in the Stable Diffusion pipeline will be fine-tuned or swap to an alternative image model with fewer false positives. Finally, for the purpose of this project, I will simplify the frontend image-loading logic by serving output files directly from the Express static route.

```
Found or updated drawing: drawing__default__1749897629145.png
100% 10/10 [00:02<00:00, 4.36it/s]
Potential NSFW content was detected in one or more images. A black image will be returned instead. Try again with a different prompt and/c
Generated and saved: generated_drawing__default__1749897629145_1749897666.png
```

Figure 12 - NSFW warning on Colab

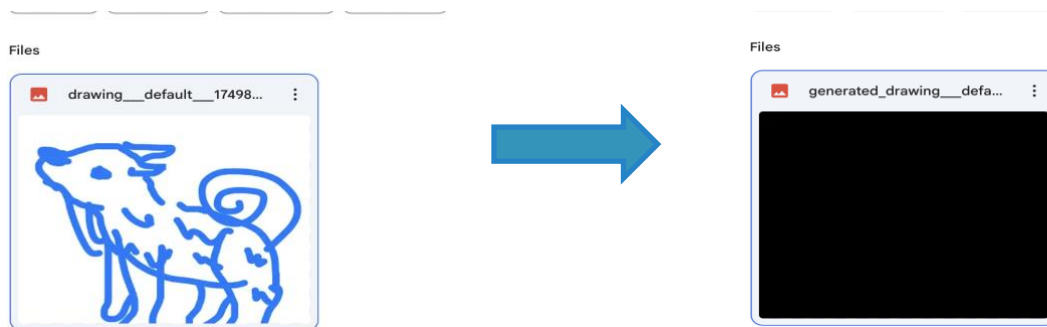


Figure 13 - Blank canvas returned after image generation

```

94 // Call Python backend to generate AI image from drawing
95 let aiImageUrl = null;
96 try {
97   const promptRaw = prompt || "default";
98   const safePrompt = promptRaw.trim().replace(/s+/g, "_").slice(0, 20);
99   const timestamp = Date.now();
100   const fileName = `drawing__${safePrompt}__${timestamp}.png`;
101   const inputPath = path.join(DRIVE_INPUT_PATH, fileName);
102
103   // Ensure input folder exists
104   fs.mkdirSync(DRIVE_INPUT_PATH, { recursive: true });
105
106   // Save uploaded image to Drive input folder
107   fs.writeFileSync(inputPath, file.buffer);
108
109   // Poll for generated image
110   const outputFileName = `generated_drawing__${safePrompt}__${timestamp}.png`;
111   const outputPath = path.join(DRIVE_OUTPUT_PATH, outputFileName);
112
113   const maxWaitTimeMs = 6000;
114   const pollIntervalMs = 5000;
115   let waited = 0;
116   //L to chat, %K to generate
117   while (!fs.existsSync(outputPath) && waited < maxWaitTimeMs) {
118     await new Promise(resolve => setTimeout(resolve, pollIntervalMs));
119     waited += pollIntervalMs;
120   }
121
122   let finalOutputFile = null;
123   const files = fs.readdirSync(DRIVE_OUTPUT_PATH);
124   for (const f of files) {
125     if (f.includes(`generated_drawing__${safePrompt}__${timestamp}`)) {
126       finalOutputFile = f;
127       break;
128     }
129   }
130 }

```

Figure 14 - Colab and Google Drive Sync

VOICE NARRATION

Currently, voice narration uses the browser's built-in Web Speech API (**Figure 15**), which meets basic functionality but sounds noticeably artificial. To create a warmer, more engaging reading experience, the next iteration will integrate a naturalistic TTS engine—such as Coqui TTS, XTTS v2, or, budget permitting, GPT-4o-mini-TTS. These models will be fine-tuned and calibrated through targeted user testing—gathering feedback on clarity, expressiveness, and emotional tone—to ensure the selected voice resonates with both children and caregivers.

```
62   const handleReadStory = () => {
63     if (window.speechSynthesis.paused) {
64       window.speechSynthesis.resume();
65       return;
66     }
67
68     if (window.speechSynthesis.spoken) {
69       return; // prevent overlapping reads
70     }
71
72     const utterance = new SpeechSynthesisUtterance(story);
73     utteranceRef.current = utterance;
74     window.speechSynthesis.speak(utterance);
75   };
76
77   const handlePauseStory = () => {
78     window.speechSynthesis.pause();
79   };
80
81   const handleStopStory = () => {
82     window.speechSynthesis.cancel();
83   };
```

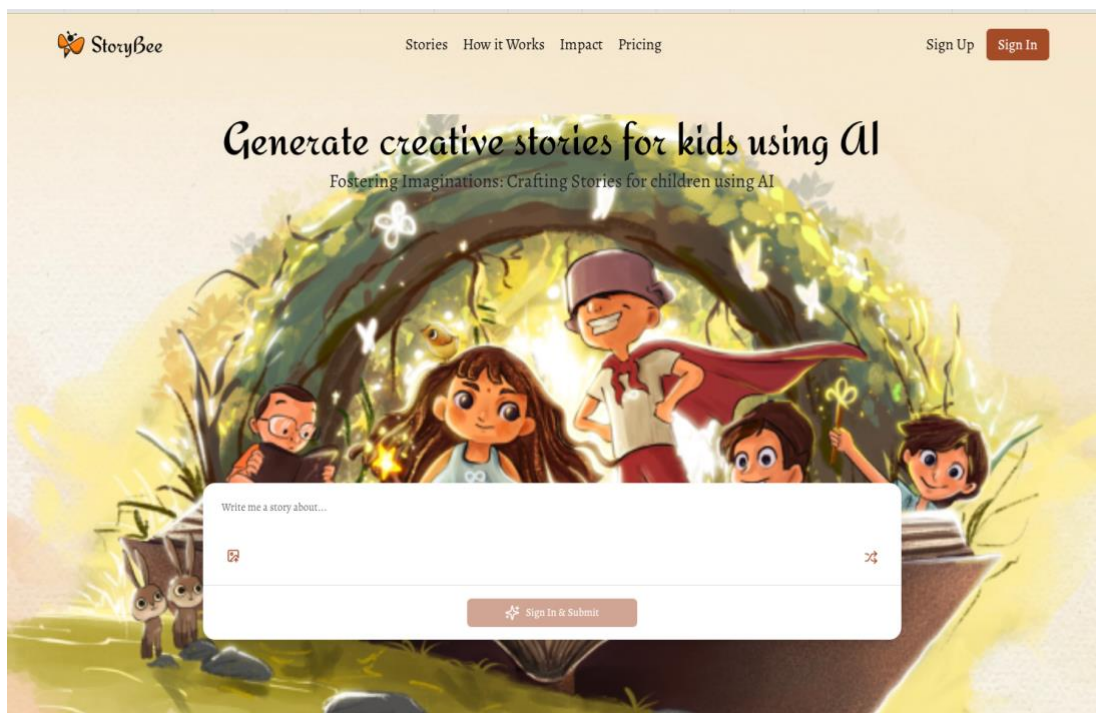
Figure 15 – Story Narration (Text-to-Speech) using speechSynthesis

APPENDICES

APPENDIX A – RELATED WORKS

A.1 STORYBEE

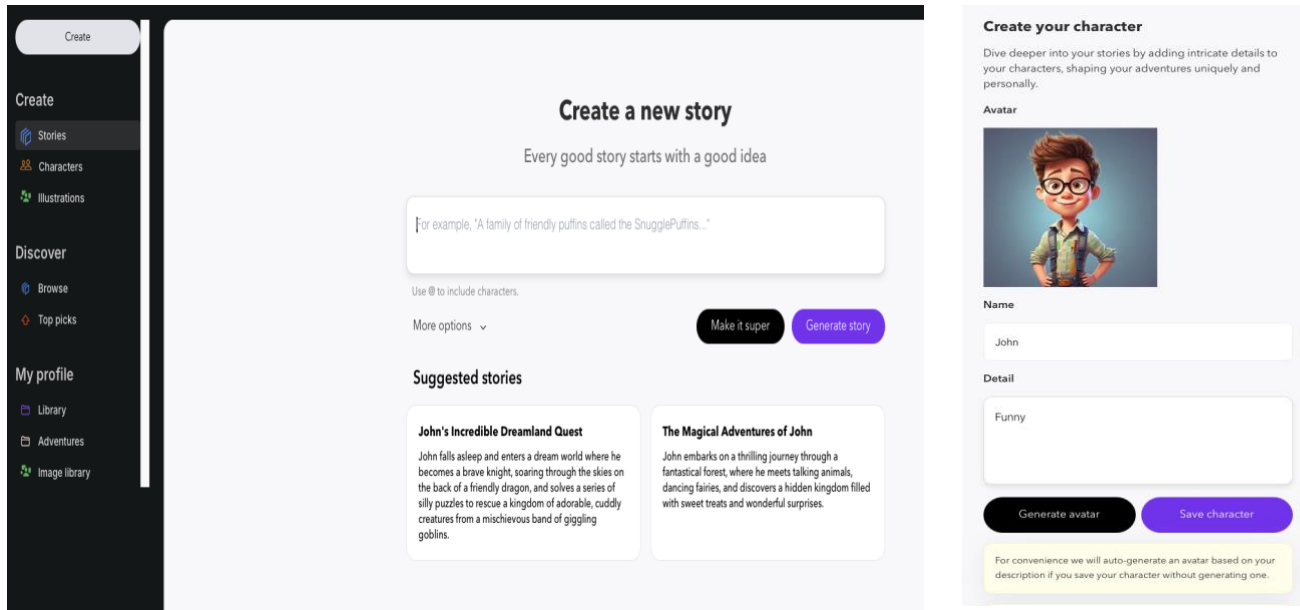
Website: <https://storybee.app>



StoryBee is an AI-powered storytelling platform that allows users to generate personalized children's stories based on input such as the child's name, desired characters, settings, or themes. Designed for parents and educators, the platform produces creative narratives within seconds, often accompanied by illustrations and optional audio narration. While intuitive and fast, StoryBee relies entirely on textual input, making it most suitable for adults or older children who can type and articulate prompts.

A.2 BEDTimestory AI

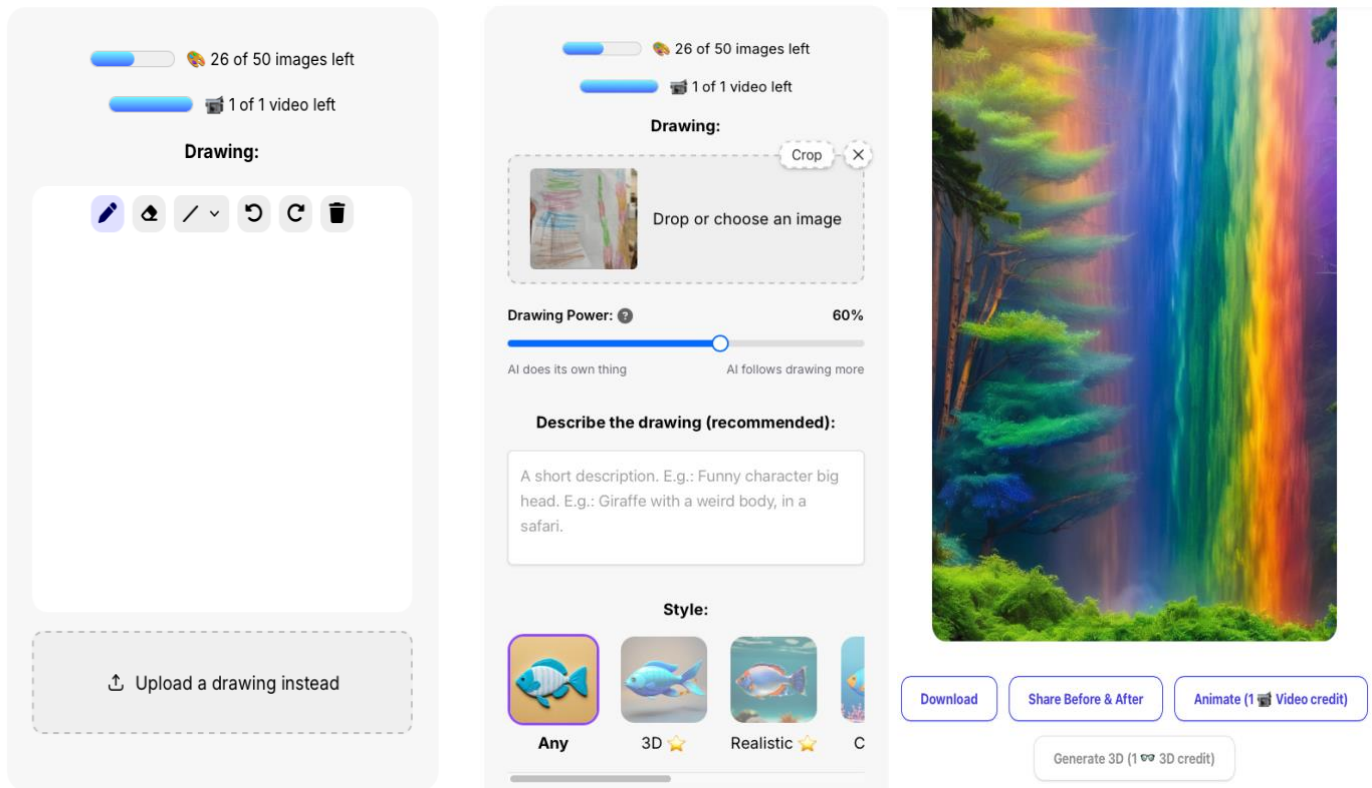
Website: <https://www.bedtimestory.ai>



Bedtimestory AI is an online platform that instantly generates personalized bedtime stories based on user-inputted keywords such as character names, themes, or morals. The interface is clear and user-friendly, allowing parents or children to quickly type in simple prompts and receive a fully-formed, engaging narrative seconds later. While it excels in speed and personalization, the system is limited to text-based interaction—it does not support image uploads, voice input, or child-driven creative assets—making it most appropriate for users comfortable with typing rather than pre-literate young children.

A.3 DRAWINGS ALIVE

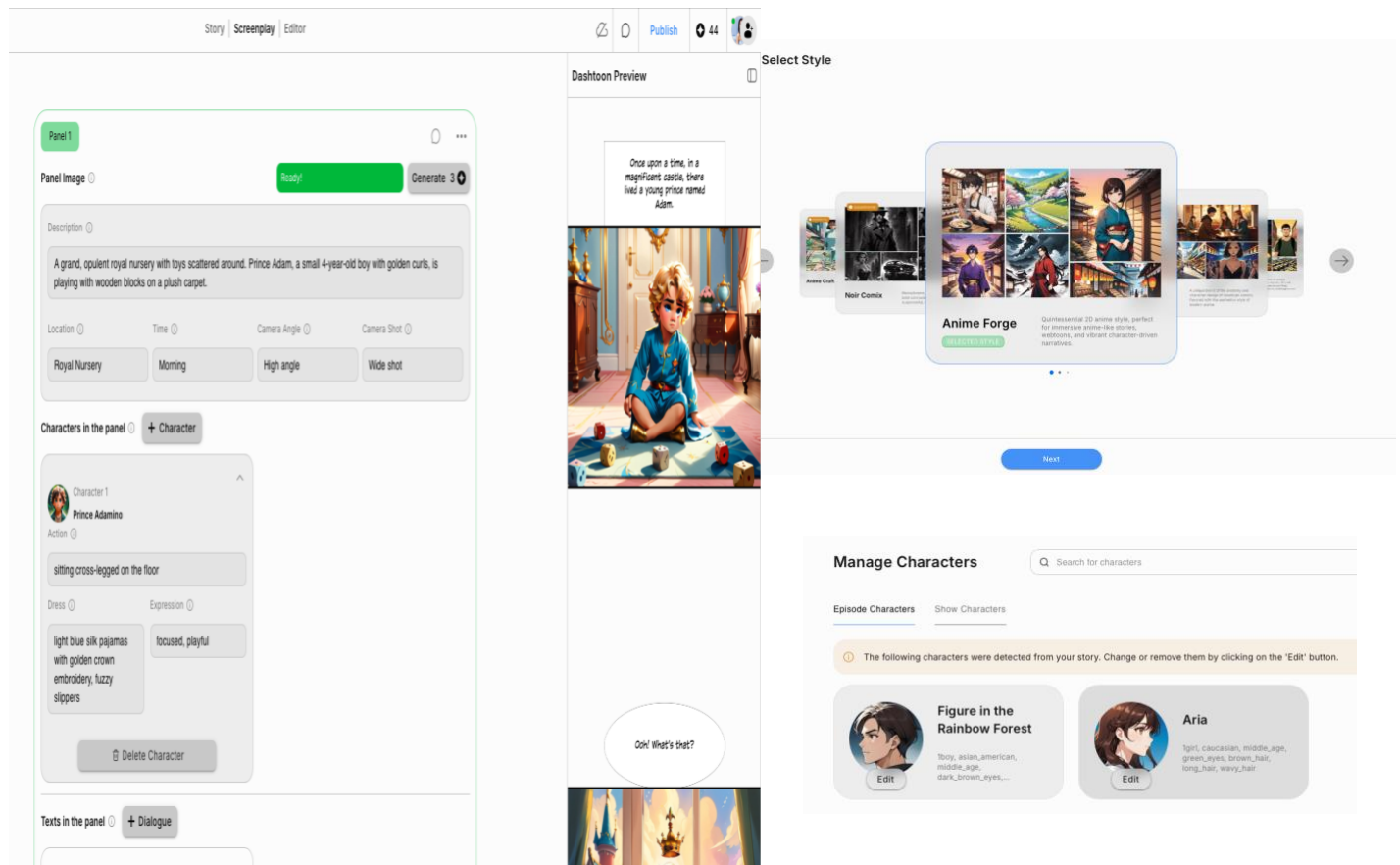
Website: <https://drawingsalive.com>



Drawings Alive is a web-based AI application that brings children's drawings to life by transforming their simple sketches into vibrant artworks, animated clips, or even 3D models. After uploading a scan or photograph of a drawing (or draw directly on the website) and optionally providing a short description, users choose from different styles—from realistic renderings to cartoons and 3D animations—and watch the artwork animate within seconds. It emphasizes visual creativity and interactive engagement, though it currently lacks storytelling features, either narrative or voice-over.

A.4 DASHTOON'S AI CHILDREN'S BOOK GENERATOR

Website: <https://dashtoon.com/ai-childrens-book-generator>



Dashtoon's AI Children's Book Generator is an online platform that quickly produces personalized children's books by letting users input themes, character details, and story elements. The system uses these inputs to generate both narrative and high-quality illustrations, delivering a complete illustrated story within minutes. It features an intuitive interface that allows users to review and customize the content before finalizing a book. However, it relies on textual prompts and does not support children's own drawings or voice inputs.

APPENDIX B – MAIN TASKS

Architectural Design (Finalize Wireframe)	• Conduct further research on UI/UX best practices for early childhood digital products. • Finalize wireframes & frontend design style.
Software Development + Functional Testing (TDD)	• Ensure the full-stack architecture is robust, with smooth and reliable communication between the React frontend and Node.js/Express backend. • Finalize the selection of pre-trained AI models for each modality, confirming that they align with the project's goals for story generation, image processing, and voice narration. • Integrate TDD method in the process of testing.
AI Models Testing & Integration	• Compare GPT-4.1 nano and MiniGPT-4 for story generation from image + prompt. • Evaluate Stable Diffusion with ControlNet for image generation in Colab. • Explore KeyBERT for keyword extraction from children's input (if necessary) • Evaluate and decide on the more natural-sounding alternative for final voice narration.
User Testing & Evaluation	• Carry out user acceptance testing (UAT) with parents, educators, and early childhood education (ECE) professionals. • Collect qualitative feedback on usability, story quality, image relevance and voice clarity.
Review & Iterations	• Iterate designs after testing and evaluation to address identified issues.
Final Report & Documentation	• Report writing and design documentation
Final Product Demo (Video)	• Generate final product demo video.

APPENDIX C – GPT-4O-MINI-TTS (TEXT-TO-SPEECH)

Website: <https://platform.openai.com/docs/guides/text-to-speech?lang=javascript>

Text to speech

[Copy page](#)

Learn how to turn text into lifelike spoken audio.

The Audio API provides a `speech` endpoint based on our `GPT-4o mini TTS (text-to-speech) model`. It comes with 11 built-in voices and can be used to:

- Narrate a written blog post
- Produce spoken audio in multiple languages
- Give realtime audio output using streaming

Here's an example of the `alloy` voice:



ⓘ Our [usage policies](#) require you to provide a clear disclosure to end users that the TTS voice they are hearing is AI-generated and not a human voice.

Quickstart

The `speech` endpoint takes three key inputs:

- 1 The `model` you're using
- 2 The `text` to be turned into audio
- 3 The `voice` that will speak the output

Here's a simple request example:

Generate spoken audio from input text

javascript

```
1 import fs from "fs";
2 import path from "path";
3 import OpenAI from "openai";
4
5 const openai = new OpenAI();
6 const speechFile = path.resolve("./speech.mp3");
7
8 const mp3 = await openai.audio.speech.create({
```

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